

Computer Logic and Digital Circuit Design (PHYS306/COSC330): Unit 4

Jordan Hanson

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Whittier College Department of Physics and Astronomy

Summary

Unit 4 Summary

1. Analog to Digital Conversion (ADC)
2. Digital to Analog Conversion (DAC)
3. Digital Signal Processing (DSP)
4. Digital Signal Processors

Analog to Digital Conversion

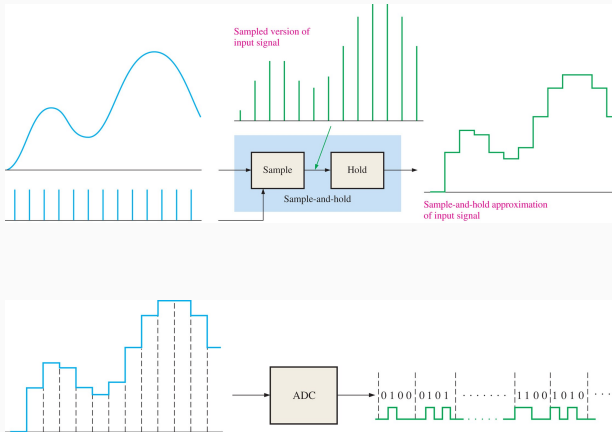


Figure 1: (Top) Analog signal being converted to digital data via sample and hold. (Bottom) Coded output.

Analog to Digital Conversion

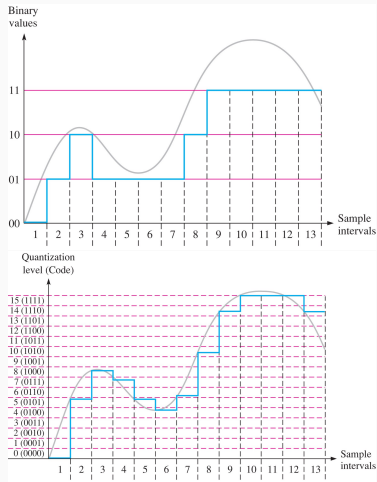


Figure 2: (Top) 3-bit conversion. (Bottom) 4-bit conversion.

Analog to Digital Conversion

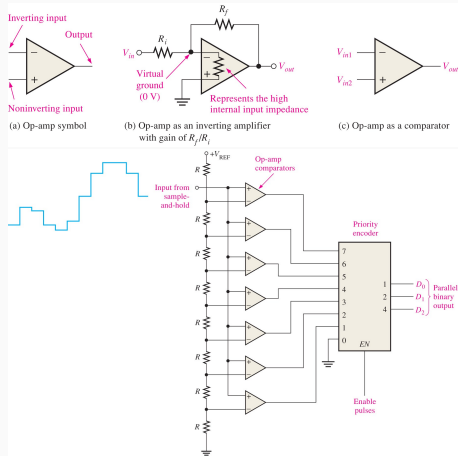


Figure 3: (Top) Op-amp comparator. (Bottom) 3-bit Flash ADC.

Analog to Digital Conversion

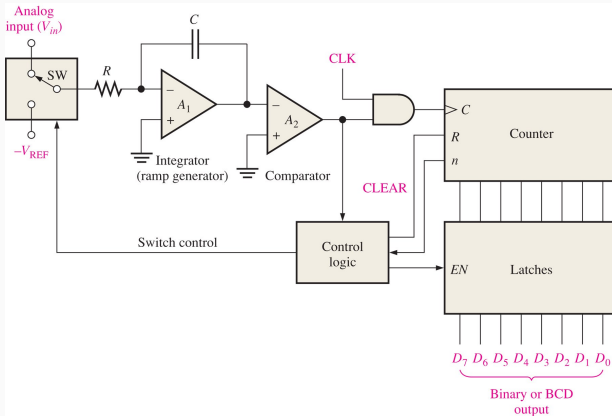


Figure 4: Dual slope ADC.

Analog to Digital Conversion

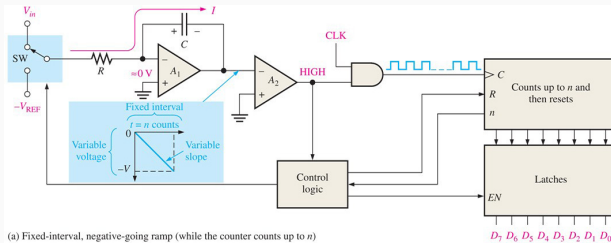


Figure 5: Dual slope ADC.

Analog to Digital Conversion

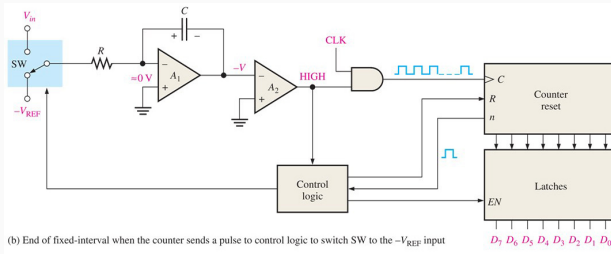


Figure 6: Dual slope ADC.

Analog to Digital Conversion

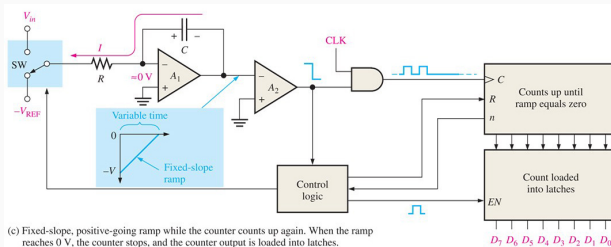


Figure 7: Dual slope ADC.

Analog to Digital Conversion

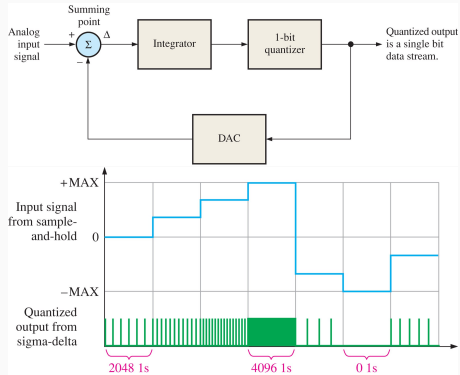


Figure 8: Sigma-Delta ADC.

Analog to Digital Conversion

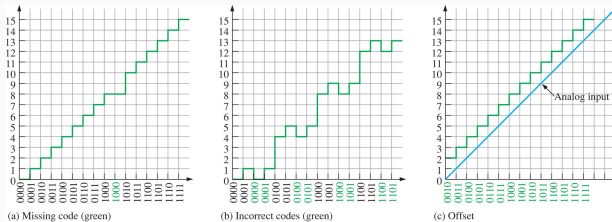


Figure 9: Types of ADC error.

Analog to Digital Conversion

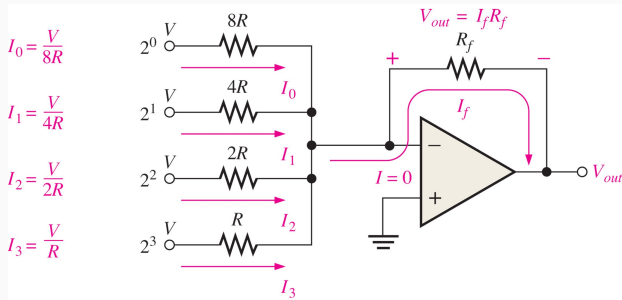


Figure 10: Binary-weighted DAC.

Analog to Digital Conversion

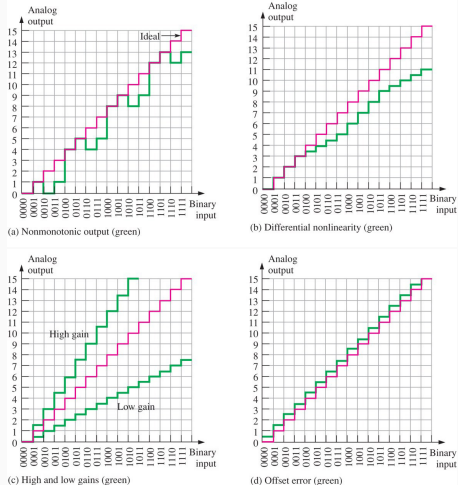


Figure 11: Types of DAC error.

Conclusion

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